

Master's Thesis Projects in 3D Vision and Geometric Deep Learning

We offer master's thesis projects in the Computer Vision Group at Chalmers, focusing on reconstructing 3D scenes from images, finding correspondences between images, and designing efficient vision transformers.

The projects build on our recent work on self-supervised 3D representations (**Poincar3** [1]), image matching (**LoMa** [2] and **RoMa v2** [3]), and efficient equivariant networks (**Flopping for FLOPs** [4]).

Project Details

Possible topics include:

- **Feedforward 3D reconstruction:** Develop accurate and efficient neural networks for predicting 3D scene structure and camera poses from images.
- **Self-supervised learning for 3D vision:** Explore how learning from unlabelled images and videos can improve reconstruction and matching.
- **Matching and reconstruction:** Improve image matching under challenging conditions and investigate how matching and 3D reconstruction can benefit each other.
- **Geometric deep learning for faster transformers:** Use geometric symmetries and equivariance to design more efficient vision transformers.

You will develop and evaluate methods together with researchers in the group. We intend to publish the results at top-tier conferences such as CVPR, ICCV or NeurIPS.

Prerequisites

We are looking for Chalmers students with **excellent grades, good programming skills and a strong mathematical foundation**. Coursework in machine learning or computer vision, and experience with Python and PyTorch, are beneficial.

Contact

Please send your CV, transcript of records and a short description of your interests to either:

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References

[1] David Nordström et al. Emergent Multi-View Geometry Through Self-Distillation. arXiv, 2026.

[Project page](#) · [Publication](#)

[2] David Nordström et al. LoMa: Local Feature Matching Revisited. ECCV, 2026.

[Project page](#) · [Publication](#)

[3] Johan Edstedt et al. RoMa v2: Harder Better Faster Denser Feature Matching. ECCV, 2026.

[Project and code](#) · [Publication](#)

[4] Georg Bökman, David Nordström and Fredrik Kahl. Flopping for FLOPs: Leveraging Equivariance for Computational Efficiency. ICML, 2025.

[Project and code](#) · [Publication](#)